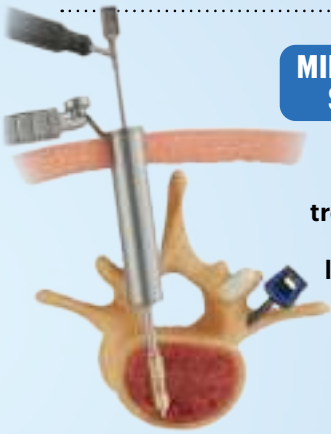
AUGMENTED REALITY NAVIGATION



Allows surgeons to see internal structures overlaid in real-time during surgery, improving safety and accuracy in complex spinal procedures

Cost: **Depends on the surgery it is used for**

MINIMALLY INVASIVE SPINE SURGERY



Uses incisions under 2 cm to treat disc problems or nerve compression, leading to less tissue damage, minimal blood loss and faster recovery

Cost: **₹1.5-5 lakh**

STEM CELL & REGENERATIVE THERAPY



Uses special repair cells in the body that can transform into different tissue types to heal degenerated spinal discs and injured nerves. Platelet-rich plasma (PRP) is also used to boost healing and ease chronic back pain

Cost: **₹3–8 lakh per treatment cycle**

THE RISK FACTORS

For many, the warning signs begin subtly—stiff shoulders after long study sessions, a nagging ache at the base of the neck or fatigue that no amount of rest seems to fix. According to Dr Ashish Tomar, senior consultant of orthopaedics & spine surgery at Faridabad's Sarvodaya Hospital, the modern lifestyle is leading to "a slow reshaping of young spines"—even in adolescence. The numbers back this. A 2024 study in the *Journal of Bodywork and Movement Therapies*, which surveyed 1,007 students in Grades 8–12 across India, found that 61 per cent of those using e-learning devices reported neck pain. For Mumbai homemaker Neha Sharma, the problem hit close to home when her 11-year-old son began struggling to tilt his head back. A visit to the paediatrician revealed an altered neck posture from months of e-learning and gaming. "He's always bent forward, whether it's for school or screens," she says. "I never imagined something like this could happen to a child." Another 2024 study in the *Journal of Pharmacy and Bioallied Sciences* found that among the 500 Indian schoolchildren surveyed, the average backpack weighed about 13.5 per cent of their body weight. Half the boys and nearly two-thirds of the girls reported back pain as a result.

Office work is equally unforgiving. In Pune, 24-year-old design intern Suresh Raghavan developed upper back pain within six months of starting a work-from-home job. The culprits: a dining chair substituting as office furniture, a laptop placed too low, and marathon work sessions without breaks. "By the end of the day, my shoulders would feel like they were made of rock," he recalls. An X-ray revealed early postural kyphosis—an abnormal forward rounding of the upper spine—linked to what is now commonly known as 'tech neck', the strain caused by constantly bending over screens. Left unchecked, his doctor warned, it could become permanent. His experience echoes the findings of a 2025 study in the *International Journal of Medical Science and Innovative Research*, which reported that half of the 300 corporate desk-job workers aged 22–40 surveyed in western India suffered from neck pain.

Obesity adds another layer of risk, overloading discs and joints, while obesity-related muscle loss and inflammation worsen pain. Nutrition, often overlooked, has a part to play too. Vitamin D deficiency softens vertebrae; magnesium shortages impair calcium absorption. In Chennai, 22-year-old architecture student Kavya Menon learned this the hard way. Living on processed snacks and avoiding outdoor activity, she was diagnosed with early osteomalacia, a condition where bones lose strength because they don't mineralise properly. "First, I thought it was just exam stress," she says, "but my bones were weaker than they should be."

Stress, say experts, is perhaps the most underestimated factor. Cross-sectional data from Indian medical undergraduates, published in the *International Journal of Basic & Clinical Pharmacology* in 2024, confirmed a high frequency of low back pain, linked to stress, sedentary habits and poor ergonomics. At AIIMS Jodhpur, for instance, three out of four students had experienced at least one episode of low back pain in the past two years, and 41.8 per cent reported it within the past four weeks. What holds true for medical students reflects the experiences of many in other high-pressure professions as well. "Chronic stress keeps neck, shoulder and lower back muscles in a constant state of tension," explains Dr Mathai. Over time, this distorts posture, compresses nerves, reduces mobility and can even trigger structural changes such as cervical spondylosis—age-related wear and tear of the neck vertebrae—and chronic lower back pain.